Strategy | **Target** A ≥ 90 | **File** 11 / 11

A+ The A+ study protocol

بروتوكول الطالب الأول — استراتيجيات الوصول إلى A+

Chapter 4 rewards a specific kind of preparation: not more hours, but the right sequence. This is what the top student in a STAT 2101 section actually does, in order.

Phase 1 — Build the skeleton (90 minutes, once)

1.1 **Write the formula sheet by hand, from memory, with the page closed.**
Then open it and mark every gap in red. Repeat two days later. Recall is what the exam measures; re-reading is not recall and feels far more productive than it is.

1.2 **Make a one-page decision tree in your own words.**
Copy the "Which rule do I use?" section above, but phrase every branch yourself. The act of rewording is what moves it into memory.

1.3 **Learn the definitions as sentences, not as feelings.**
Sample space, outcome, event, simple, compound, mutually exclusive, independent, dependent, complement. These are CLO1 marks; they appear on quizzes verbatim and take one minute each to secure.

Phase 2 — Drill by question type, not by page order

2.1 **Group every exercise by which rule solves it**
before solving any of them. Fifteen minutes of sorting teaches you more about rule selection than two hours of grinding, because rule selection — not arithmetic — is where marks are lost.

2.2 **Do the assigned exercises properly: p. 247, numbers 6, 8, 16 and 40.**

These are named in the course outline, which means the tutor has them, which means variants of them are the most likely quiz questions in the course. Then do the Moodle exercises — they are listed for both Week 5 and Week 6.

2.3 **Solve every "Self-study" example in the slide deck.**

Examples 4-13, 4-21, 4-22, 4-26, 4-28, 4-40, 4-45, 4-54 and the three review exercises are marked self-study, which means they are *not* done in class and are therefore exactly where an examiner finds unfamiliar-looking material. All of them are solved in the lesson files of this series — cover the solution and redo them from the question alone.

2.4 **Build the "twin" of each problem you solve.**

Did it say *with* replacement? Redo it *without*. Did it ask for "exactly 2"? Also do "at least 1". Twin-building is the single highest-yield habit in this chapter because the examiner builds twins for exactly the same reason.

Phase 3 — Write like a marker reads

3.1 **Open every answer by defining your events in symbols.**

"Let N = the staff person is a nurse, M = the staff person is male." Two lines, and the rest of the solution writes itself — and the marker can follow you even if the arithmetic slips.

3.2 **Write the sample space in set notation whenever it is small enough.**

The lecturer flags this explicitly on the Example 4-6 slide: "Sample spaces and events need to be expressed using set notations." That is a marking instruction in disguise.

3.3 **Name the rule before you use it.**

"Since the events are not mutually exclusive, Addition Rule 2 applies." One sentence, often one mark, and it forces you to check your own choice.

3.4 **Finish with a sentence in words.**

"So there is about a 54% chance that at least one circuit is defective." Interpretation is a stated outcome of this course (CLO8) and costs five seconds.

Phase 4 — The exam hour itself

4.1 **Brain-dump the formula sheet onto the back page in the first 90 seconds**, before reading any question. You are then calculating from a written reference instead of from a memory under pressure.

4.2 **Underline the decisive words as you read:** *and, or, not, at least, at most, given, without replacement, order is important*. Every one of them selects a formula. The question is telling you which rule to use; most lost marks come from not listening.

4.3 **Run the three checks on every numerical answer:** is it between 0 and 1? do the complementary parts sum to 1? does the magnitude feel right (a rare event should come out small)? These catch nearly every slip.

4.4 **Budget by marks, not by difficulty.** The midterm is 60 minutes across Chapters 1–4. Chapter 4 questions look longest but have the most structured method — do them with a timer at home so you know your real pace before exam night.

استراتيجية الطالب الأول — باختصار

- **اكتب ورقة القوانين من ذاكرتك** ثم صحّحها. الاسترجاع هو ما يُختبر، وليس القراءة.
- **صنّف التمارين حسب القانون المستخدم** قبل حلّها — ضياع الدرجات يكون في اختيار القانون لا في الحساب.
- **حلّ كل الأمثلة المكتوب عليها Self-study** فهي غير محلولة في المحاضرة، ومنها تأتي الأسئلة غير المألوفة.
- **اصنع "التوأم" لكل مسألة:** إذا كانت with replacement فأعدها without، وإذا كانت "exactly 2" فأعدها "at least 1".
- **في الامتحان:** اكتب القوانين في أول 90 ثانية، ضع خطًا تحت الكلمات المفتاحية (and / or / not / at least / given / without replacement)، وتأكد أن كل إجابة بين 0 و 1.
- **اكتب جملة تفسيرية في النهاية** — "أي أن احتمال وجود قطعة معيبة واحدة على الأقل يقارب 54%". هذه درجات شبه مجانية.

The fastest wins, ranked

RANK	ACTION	TIME COST
1	Memorise the five phrase translations (exactly / less than / at most / more than / at least).	10 minutes
2	Memorise `P(at least one) = 1 − P(none)` and scan every question for "at least".	5 minutes
3	Drill the permutation-vs-combination decision on 20 one-line prompts.	25 minutes
4	Practise building a two-way table from a word problem (Example 4-21 is the template).	20 minutes
5	Learn the definitions verbatim for CLO1 marks.	20 minutes
6	Rehearse the Casio keystrokes for `x!` , `nPr` , `nCr` until they are automatic.	10 minutes

That is ninety minutes of work that covers the majority of the marks available in Chapter 4. Everything else is practice volume.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability

04 · Lesson 4-4 — Counting Rules

05 · Lesson 4-5 — Probability and Counting Rules

06 · Master formula sheet

07 · Which rule do I use?

08 · The 14 traps that cost marks

09 · Review exercises, solved

10 · The A+ study protocol ← this file

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.